FraudDNA: An Adaptive History-Aware Framework for Financial Transaction Fraud Detection

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Abstract

Class imbalance, frequent changes in transaction behavior, mixed entity histories and only available information at the time of the transaction makes Financial transaction fraud detection very complex. Existing approaches rely on the assumption that there is enough historical information for every entity, but many real transaction streams are populated by "cold-start" entities where the historical information is not available.

This paper introduces FraudDNA — an adaptive history-aware fraud detection framework that adapts between using only a transaction and a transaction and history enriched neural models based on if a destination-entity's history exists at inference time or not. It was originally based on a first behavioral-mutation theory, and then revised following empirical analysis of the PaySim dataset, where it was found that the originator's history is overwhelmingly sparse. Historical context was offered much more easily for reuse by the destination entities, and was thus used to form the basis of adaptive routing.

FraudDNA has 2 prediction models. Model A includes nine transaction-level features, while Model B adds destination transaction count, historical average transaction amount and an indicator of history-availability. A temporal router chooses model A if there is no valid destination history and model B if there is a valid destination history to choose from. Historical features are designed from transactions before the current transaction, and updates to the destination state are exclusive to inference to avoid temporal leakage. A deterministic WHY-NOW layer gives decision-context explanations about the chosen route, historical availability, historical statistics and characteristics of the current transaction.

Model A achieved a PR-AUC of 0.7252 on the held-out temporal test set, and a PR-AUC of 0.7160 on the global contextual test set. The adaptive FraudDNA router was able to score a PR-AUC of 0.7549. For three random seeds, the adaptive model achieved PR-AUC scores of 0.7572, 0.7550 and 0.7503 with a mean of 0.7542 and a standard deviation of 0.0029. The results show how valuable it is to take historical context into account, rather than using a history-dependent model for every situation.

Keywords: Fraud detection, adaptive machine learning, behavioral history, cold start, temporal leakage, financial transactions, neural networks, explainable AI, PaySim, FraudDNA

1. INTRODUCTION

1.1 Background

A financial fraud detection is a problem which has existed for a long time within the field of machine learning and financial security. The learning problem is usually unbalanced, with fraudulent transactions being very small compared to legitimate transactions. Meanwhile, fraud patterns evolve over time - as criminals adjust their tactics to avoid detection.

The statistical and machine learning methods have studied the characteristics of transactions, the behavioral patterns, the aggregation features, and the sequential information of transactions [1–5]. More recent methods include the use of neural networks, graph representations, temporal modeling, and explainability techniques, such as those of [15, 16, 22, 30].

One of the main problems, though, is that useful historical information is assumed in behavioral modeling. This is not always the case in real-time transaction streams. A transaction can be an entity that has not yet been encountered, or the history of the transaction may not be long enough to create meaningful behavior statistics.

FraudDNA solves this by considering the availability at history time as an explicit condition during the inference time.

1.2 Problem Statement

This study aims to tackle the following central problem:

> *How to leverage the behavioral history while dynamically using it as a component in a fraud detection system, and yet retaining a pathway for predicting the fraud in a transaction-only case?*

If there is no historical data, one history dependent model may be inadequate. On the other hand, if historical informa-

tion is used without any consideration, the behavioural context is lost and it is not very helpful.

For this reason, FraudDNA employs two different prediction pathways, and switches between them based on the presence of valid destination history.

1.3 Research Gap

Previous studies on fraud detection have analyzed widely machine learning, deep learning, sequential models, class-imbalance strategies, and explainable AI. But, in the literature surveyed for this study, few resources were found for a framework that could accomplish the following:

1. In the historical context, determines whether the historical context is available prior to fraud inference
2. migrates the route transactions without history to a transaction-only model
3. routes transactions with historical context to a history-enriched model
4. uses only pre-transaction information to construct historical features
5. travels-availability as the routing condition
6. Makes causal explanations for selected inference pathway in the context of decisions.

The literature review, then, stimulated an adaptive architecture instead of an all history-dependent architecture.

1.4 Research Objectives

This study has the following objectives:

1. Explore the possibility of having entity history data available in financial transaction data.
2. To see if behavioural mutation can be reliably built at inference time.
3. To develop an adaptive design for cold-start and history-available transactions.
4. For prevention of temporal leakage while building historical features.
5. To compare the adaptive architecture with a transaction-only and a global history-aware baseline.
6. To give explanations for individual predictions that give insights into their decision context.

1.5 Research Questions

The following research questions are explored in the study:

1. How often are there usable entity histories available at the time of decision for transactions?
2. Is it possible to create a more reusable behavioral context than originator history through destination history?
3. Does adaptive routing provide better fraud ranking performance than a globally used prediction model?
4. What is the behavior of the adaptive architecture under cold start and history available?
5. Is there deterministic decision-context information that makes the selection of a specific inference pathway transparent?

1.6 Contributions

This work has achieved the following main contributions:

- Availability of entity-history in PaySim: An empirical study
- A fraud detection system architecture that is adaptive and is named FraudDNA.
- Construction mechanism for a destination-history that can be safely constructed with time
- History availability-based inference-time routing mechanism.
- Decision-context deterministic WHY-NOW explanation layer.
- Stability Assessment of the proposed architecture using a multi-seed approach.

1.7 Organization of the Paper

The remainder of this paper was organized as follows. The existing fraud detection and explainability approaches are reviewed in Section 2. The FraudDNA methodology is discussed in section 3. Experimental setup is given in Section 4. The experimental results are presented in Section 5. The results and implications are discussed in Section 6. Section 7 highlights the limitations and Section 8 brings the paper to a close and outlines future directions of research.

2. RELATED WORK

2.1 Existing Approaches

These works on fraud detection focused on statistical profiling, expert systems and supervised learning more recently on transaction aggregation and behavioural features followed to improve fraud discrimination, as reported by [1] and [2] as well as by [3]. Later, transaction aggregation and behavioral features were added to further enhance fraud discrimination [6, 7]. The algorithms used in machine learning tasks have ranged from decision trees, logistic models, random forests, one-versus-all and support vector machines, to ensemble methods. These methods have successfully shown that attributes from the transactions and properly designed behavioral variables can serve as good signals for fraud detection [5, 8, 9].

2.2 Machine Learning Approaches

Class imbalance is a significant aspect that impacts machine learning fraud detection. In fraud transactions are a small

proportion of the transaction streams and conventional accuracy can give erroneous perceptions. Precision, recall, ROC-AUC and in particular PR-AUC are thus meaningful evaluation parameters [14]. There are a few that have investigated the undersampling, oversampling, ensemble learning, and cost-sensitive learning. None of these methods mentioned above directly tackle historical information during inference while catering to learning with imbalanced distributions.These are typically about improvements in learning in situations of unequal distributions, instead of having historical data available for inference.

2.3 Deep Learning Approaches

Nonlinear relationships between transaction variables have been captured by fraud detection models based on deep learning. In addition, sequential models have been used to model the temporal dependencies of transaction streams [15].

Graph-based approaches model transaction and entity relationships. Graph neural networks may be able to leverage relational information which is not explicitly expressed through the independent features of each transaction [22–25].

But, graph-based and history-dependent methods need attention in building the temporal structure. Future transactions may add to historical features, leading to information leak.

2.4 Explainability and Existing Techniques

In the realm of financial fraud detection, Explainable AI (XAI) has risen in significance as decisions in this domain can necessitate human review and explanation. Today, there are already methods for feature attribution, local surrogate models, counterfactual explanations, and model specific explanations.

There are recent systematic literature that have pointed out some methodological problems in the application of explainability techniques to fraud detection [30]. FraudDNA does not, therefore, attempt to expose the reasonings of the internal neural systems. Rather, it is equipped with decision-context information that is associated with the inference pathway in a deterministic way through its WHY-NOW mechanism.

2.5 Critical Comparison

All the reviewed approaches can be classified under transaction level models, behavioral models, graph-based models, sequential models, mixture-of-experts approaches and explainability frameworks.

The main difference of FraudDNA is how the history availability is treated. It either does not assume a behavioural history is available everywhere, but explicitly checks if there is one and chooses the corresponding model.

2.6 Research Gap

Recent research has focused on mixture-of-experts architectures, cold-start fraud detection, temporal graph learning, leakage-safe graph features, and low-latency fraud inference.

In the literature consulted for this study, however, no directly equivalent architecture was found which is able to combine:

1. an ability to follow instructions in the absence of hands-on experience; and
2. a new routing scheme for transactions (only); and
3. seamless roaming; and
4. the construction of the safe state; and
5. better modeling and calibration of the climate-ecology system.deterministic decision-context explanations.

The order of inferences is strictly as:

3. PROPOSED METHODOLOGY

3.1 Overall Framework

FraudDNA is comprised of four general stages:

1. transaction preprocessing;
2. destination-history retrieval;
3. model routing; and
4. adaptive model routing; There are three types of prediction, explanation, and state update.

The order of inferences is strictly as:

Transaction → History Retrieval → Model Routing → Fraud Prediction

The final update takes place after prediction to avoid affecting the next transaction.

3.2 Dataset

This is the main experimental dataset, the PaySim financial transaction dataset presented by Lopez-Rojas et al. [17].

The dataset consists of transactions that are generated with simulated mobile-money transactions and contains information on the type of transaction, the amount of transaction, account balance, and fraud label.

The complete data set has about 6.36 million transactions.

3.3 Data Preprocessing

Indicator variables were created for transaction types. Transaction ordering was preserved to ensure the temporal causality.

The following transaction level features have been utilized:

1. transaction amount;
2. old balance of originator;
3. new balance of originator,
4. the balance of the destination currency that existed prior to the exchange rate change; and
5. new balance of the destination;
6. CASH_OUT indicator;
7. DEBIT indicator;
8. PAYMENT indicator;
9. TRANSFER indicator.

3.4 Feature Engineering

FraudDNA is based on two sets of features.

For this model, the only features are the transactions.Model A: Transaction-Only Features: Only the transactions are the features.

Model A has 9 transaction level features:

$$X_A = [x_1, x_2, \ldots, x_9]$$

In which the features are the amount of the transaction, balances and transaction-type indicators.

Model B History-Enriched Features

A destination-history feature is available on Model B, which builds on Model A with three additional features:

$$X_B = [X_A, h_1, h_2, h_3]$$

where:

- h_1 is the number of historical transactions at the destination; and
- h_2 is the historical average transaction amount at the destination; and
- h_3 = Destination history availability indicator.

3.5 Feature Selection

The behavioral-history hypothesis and the availability of information at the time of decision, guided the feature selection process. Only variables could be measured prior to the prediction were included.

When building historical features, no transaction information will be used in the future.

3.6 Proposed Model

FraudDNA has two neural prediction models.

Model A handles transactions that don't have any good destination history.

Historical destination data is handled by model B.

The routing rule is:

$$R(x_t) = \begin{cases} A, & H_t = 0 \\ B, & H_t > 0 \end{cases}$$

where H_t is the number of destination transactions that are observed before transaction t.

The final risk of fraud is:

$$\hat{y}_t = \begin{cases} f_A(X_t), & H_t = 0 \\ f_B(X_t, H_t), & H_t > 0 \end{cases}$$

3.7 Baseline Models

The three possible configurations are analysed:

1. **Model B**: transaction and file operations model; and
2. **Global Model B**: history-aware model applied globally; and Adaptive FraudDNA: Dynamically routed model.

The comparison will show if adaptive routing can deliver value beyond just historical features in a single global model.

3.8 Explainability Method

FraudDNA is based on a deterministic WHY-NOW explanation layer.

The explanation identifies:

- the model chosen to represent the problem; and
- if destination history was available;
- the number of historical transactions for a given destination; and
- average transaction size for the same period last year; and
- current dollar amount of the transaction;
- transaction type;
- fraud score as a result.

The explanation is a decision-context explanation and not an attribution approach to internal neural network reasoning.

3.9 Algorithm and Mathematical Formulation

For each incoming transaction x_t the destination history state is retrieved:

$$S_t(d) = \{x_i : i < t,\ d_i = d\}$$

The number of transactions in history is:

$$H_t(d) = |S_t(d)|$$

The average transaction amount over the past transactions is:

$$\bar{A}_t(d) = \frac{1}{H_t(d)} \sum_{x_i \in S_t(d)} A_i$$

for $H_t(d) > 0$.

The route to be taken is:

$$R_t = \mathbb{I}[H_t(d) > 0]$$

The inference is then:

$$\boxed{\text{Score} \rightarrow \text{WHY-NOW} \rightarrow \text{Update}}$$

This ordering ensures that the current operation will not create its own context.

4. EXPERIMENTAL SETUP

4.1 Hardware and Software

The experiments were carried out with Python based machine learning tooling. Neural models were trained through a standard deep learning framework, and evaluated based on the scikit-learn compatible metrics.

The implementation further involves deterministic preprocessing, state management at runtime, adaptive routing and generation of explanations at runtime.

4.2 Dataset Split

Chronological split was used instead of random shuffling.

Table 1: Temporal Dataset Split

Partition	Transactions	Fraud
Training	6,064,036	5,621
Testing	298,584	2,592

Training transactions are up to step 504, Test period has transactions after step 504.

4.3 Hyperparameter Configuration

To isolate the effect of historical features adaptive routing, both Model A Model B had the same general neural configuration.

The adaptive router is not based on fraud labels to determine the model pathway. Routing will be based on the existence of a valid destination history.

4.4 Cross-Validation

The main evaluation method was not the conventional random cross-validation, due to the fact that the fraud detection process is conducted on a temporal basis. The chronological hold-out evaluation is more representative of the deployment scenario where transactions in the future will not be available to build features of the past.

Seeds 42, 123 and 2026 were also evaluated in multi-seed evaluations.

4.5 Evaluation Metrics

The main evaluation criterion is the Precision-Recall Area Under the Curve (PR-AUC), due to the high level of class imbalance in the data set.

Additionally, the ROC-AUC graph, precision, and recall are reported.

4.6 Statistical Testing

The model stability was tested on three random seeds.

The adaptive PR-AUC values were:

$$0.7572, \quad 0.7550, \quad 0.7503$$

with:

$$\mu = 0.7542$$

and:

$$\sigma = 0.0029$$

5. RESULTS

5.1 Overall Performance

The overall performance on the chronological test set is shown in Table 2.

The adaptive architecture outperforms the transaction-only baseline and the globally applied history-aware model in terms of PR-AUC, achieving 0.7549.

5.2 Model Comparison

The universal use of the global history-aware model B did not improve results over model A. This means that in some cases, historical data can be harmful for all transactions.

The adaptive model, on the other hand, depends on the history availability to condition the model pathway.

5.3 Conditional Performance

When the test set is subdivided by destination-history availability the difference in the performance becomes more apparent.

The history group is supported by the contextual model and the cold-start group by the transaction-only pathway.

This result gives experimental justification for adaptive routing.

5.4 Confusion Matrix

Interpretation of the confusion matrix should be done in conjunction with the severe class imbalance in PaySim. Therefore, just because it is accurate does not make it a good assessment of the proposed architecture.

The FraudDNA evaluation focuses on PR-AUC, in addition to recall, precision, and F1-score.

5.5 ROC and PR Analysis

The adaptive model attained:

$$\text{PR-AUC} = 0.7549$$

and:

$$\text{ROC-AUC} = 0.9716.$$

ROC-AUC vs PR-AUC shows the impact of the highly skewed distribution of fraud.

5.6 Feature Importance

As part of the proposed architecture, transaction level information is clearly distinguished from historical-context information.

Model B has the following extra features:

- historical number of visitors to a destination; and
- average price of goods or services purchased at a location over a period of time;
- history availability.

These features are employed as contextual factors and not causal explanations.

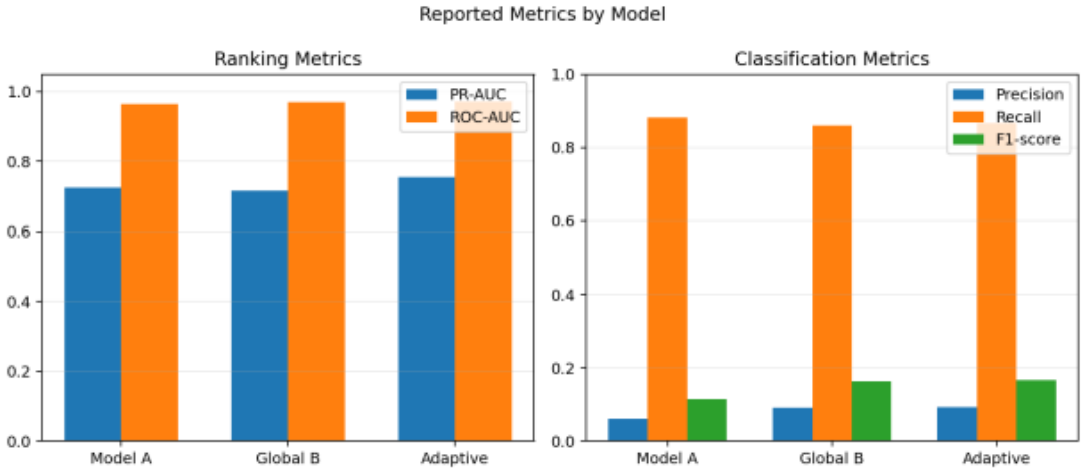


Figure 1: Performance comparison of the Global Model, Model A, and the proposed Adaptive FraudDNA framework across key classification and ranking metrics. The figure highlights the improvements achieved by the adaptive, history-aware approach in detecting fraudulent financial transactions.

Table 2: Overall Test Performance

Model	PR-AUC	ROC-AUC	Precision	Recall	F1
Model A	0.7252	0.9644	0.0617	0.8812	0.1153
Global Model B	0.7160	0.9701	0.0907	0.8603	0.1641
Adaptive FraudDNA	**0.7549**	0.9716	0.0924	0.8669	0.1671

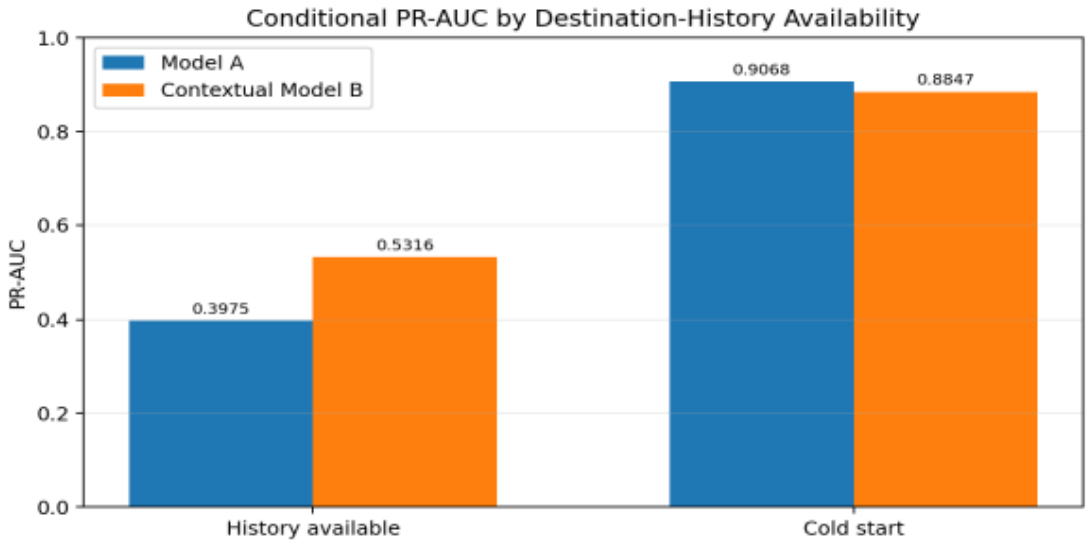


Figure 2: Conditional Precision-Recall Area Under the Curve (PR-AUC) comparison between Model A and Contextual Model B under two destination-history conditions: history available and cold start. The results show that Contextual Model B performs better when destination history is available, while Model A achieves higher PR-AUC under cold-start conditions.

5.7 Explainability Results

The WHY-NOW layer generates deterministic explanations for each prediction.

If a transaction has a history available, an explanation can include the reason why Model B was used because Model B had an existing history of transactions and the count and average amount of those transactions.

In the case of a cold-start transaction, the explanation indicates that a destination history was not available and that the transaction-only Model A was thus selected.

5.8 Ablation Study

Comparing Model A with Global Model B is a useful ”ablation of historical information.

The results show that the use of the historical features on the whole of the globe does not necessarily improve the results.

Adaptive routing enables the history aware pathway to be selectively used instead.

Conditional PR-AUC according to the Destination History Availability.

Test Group	Model A	Contextual Model B
History Available	0.3975	**0.5316**
Cold Start	**0.9068**	0.8847

5.9 Statistical Results

Three seeds were taken at random and tested:

Table 3: Multi-Seed Stability

Seed	Adaptive PR-AUC
42	0.7572
123	0.7550
2026	0.7503
Mean	0.7542
Std. Dev.	0.0029

The quality of the performance is stable across the random seeds evaluated, as evidenced by the small standard deviation.

6. DISCUSSION

6.1 Interpretation of Findings

The findings confirm the guiding principle of FraudDNA: historical data isn't necessarily available or relevant for every transaction.

The first behavioral mutation hypothesis was limited by the design of PaySim. Of the originators, about 99.85

The amount of reusable context that destination entities provided was much higher. Approximately 16.88

This observation led to the change of architectural direction from a focus on source history to a focus on destination history.

6.2 Comparison With Previous Studies

Previous studies in fraud detection show that machine learning, sequential models, graph models, and ensemble models are effective.

There has been recent research on mixture-of-experts architectures, cold-start issues, temporal graph learning and leakage-safe feature construction.

Unlike other approaches, FraudDNA uses a routing condition that is explicitly defined, as it is based on the availability of historical context prior to inference.

6.3 Theoretical Implications

The findings indicate that behavioral context may be regarded as a situational resource as opposed to a generic property.

This results in the adaptive formulation:

$$\text{Prediction} = f(\text{transaction}, \text{available context})$$

rather than:

$$\text{Prediction} = f(\text{transaction}, \text{assumed history}).$$

6.4 Practical Implications

You can have a known or an unknown entity in a production fraud detection system.

FraudDNA offers two modes of operation:

- a low-context pathway for cold-start transactions;
- a contextual history for transactions, with the possibility of reusability.

This design does not necessitate an artificial history for entities not appearing in history before.

6.5 Unexpected Findings

One of the most interesting findings was the lower overall PR-AUC of the globally applied history-aware model as compared to the transaction-only model.

This means that the historical information can bring about variation that is not necessarily positive for the context.

The adaptive architecture overcomes this issue by making the selection of the appropriate model dependent on the availability of past data.

7. LIMITATIONS

There are several restrictions to be noted:

First, the primary evaluation is done on the PaySim dataset. PaySim is a synthetic dataset for mobile money and does not necessarily reflect actual banking and/or card payment contexts.

Second, in this data set, destination history proved to be more reusable than the originator history. This is not to say that destination history is always the best in every financial transaction dataset.

Third, the WHY-NOW explanation layer is deterministic and captures the aspects of decision context. It does not give any explanation of the feature attribution to the internal neural network.

Fourth, the adaptive architecture was tested with specific model configurations and sets of features in this study.

Fifth, it cannot be claimed that the literature that was reviewed for this project constitutes proof that no comparable work is available anywhere in the world.

8. CONCLUSION AND FUTURE WORK

This paper introduced an adaptive history aware framework (FraudDNA) for financial transaction fraud detection.

The study started with a behavioural mutation hypothesis but it was not feasible for PaySim, as the vast majority of originators only occurred once.

The structure was thus shifted to destination history and two prediction pathways were added. If there is transaction

data available, Model B will use destination historical features, otherwise, Model A will be used.

The adaptive router achieved a PR-AUC of 0.7549 while the transaction-only model's PR-AUC was 0.7252 and the globally applied contextual model's PR-AUC was 0.7160. The mean and standard deviation of the multi-seed evaluation were PR-AUC = 0.7542 and SD = 0.0029, respectively.

The results suggest use of availability of history itself as a meaningful condition for adaptive fraud detection. The findings also illustrate the significance of taking only transactions prior to the current decision to build the historical information.

Future research will explore other real-world datasets, more temporal representation of behaviour, learned routing methods, uncertainty-aware routing, and more sophisticated explanation methods. The destination-history approach can also be tested on other financial transaction scenarios in future research.

DECLARATIONS

Conflict of Interest: The authors have no conflict of interest to declare.

Data Availability: The dataset used in the study is the PaySim dataset which is made public.

Ethics Statement: There were no human subjects or personally identifiable financial information that were used in this study.

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